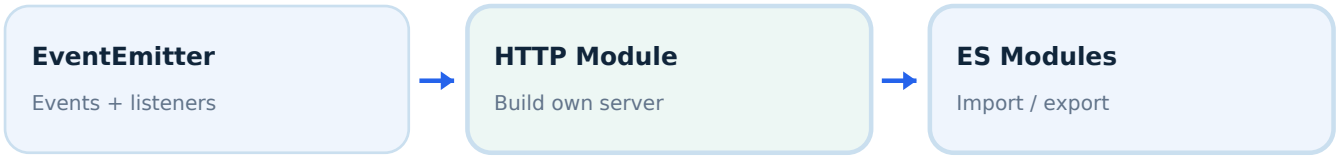
LEARNING JOURNAL / 20 OCTOBER 2026

Node.js Recent Topics Illustrated Complete Notes

Focus: aaj ke ES Modules, aur recently completed HTTP Module aur EventEmitter. Har topic mein examples, diagrams aur woh mistakes shamil hain jo practice mein saamne aayi thin.



56/90	50/70	30/40
EventEmitter practice	HTTP server practice	ES Modules tasks

Aaj ka key point

ES Modules mein named export ke liye { } aur default export ke liye import karte waqt without braces syntax use hoti hai. Import path ko exact file name / extension se match karo.

Tumne kya seekha?

ESM: import, export, export default, named + default together, file extensions, function return values.

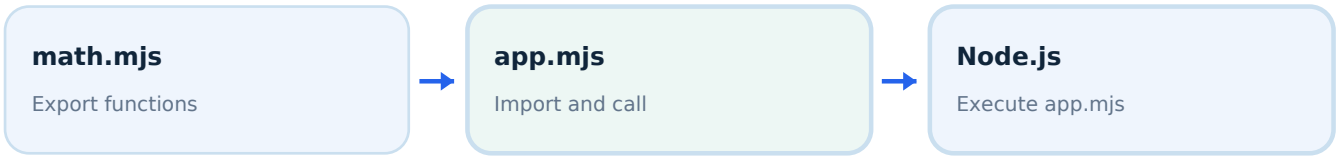
HTTP: createServer, req.url, res, port, Content-Type, res.end, statusCode=404.

Events: EventEmitter, on, once, emit, off, function references.

Note: Practice scores historical attempts hain; corrections ke baad concepts revise ho chuke hain.

01 | ES Modules - Basics

ES Modules (ESM) standard JavaScript module system hai. Code ko alag files mein rakh kar export se share aur import se use karte hain.



ES Modules enable karne ke 2 tareeqay

Option	Kya karna hai
.mjs files	math.mjs aur app.mjs banao; Node.js automatically ESM samjhta hai.
package.json	"type": "module" rakho; phir .js files ko ESM treat kiya jata hai.

Named export aur named import

Ek file se multiple named functions export kar sakte ho. Import mein names exact match hone chahiye (jab tak alias na do).

```
// math.mjs
function add(a, b) { return a + b; }
function subtract(a, b) { return a - b; }
export { add, subtract };

// app.mjs
import { add, subtract } from "./math.mjs";
console.log("Addition:", add(10, 5));       // 15
console.log("Subtraction:", subtract(10, 5)); // 5
```

> **Important: file path**
> Agar file ka naam math.mjs hai, to "./math.mjs" import karo - "./math.js" likhoge to file na hone par ERR_MODULE_NOT_FOUND ho sakta hai.

CommonJS se difference

CommonJS	ES Modules
module.exports / require()	export / import
Typically .cjs or CommonJS .js	.mjs ya "type": "module" wali .js
const x = require("./x")	import x from "./x.mjs"

02 | Default + Named Export

Default export: ek module mein maximum ek default export. Import karte waqt braces use nahi hoti, aur import ka local naam change kar sakte ho.

```
// welcome.mjs
export default function greetStudent(name) {
  return `Welcome ${name} to Node.js Learning!`;
}

// app.mjs
import greetStudent from "./welcome.mjs";
console.log(greetStudent("Junaid"));
console.log(greetStudent("Ali"));
```

Default + named export same file mein

```
// result.mjs
export default function studentInfo(name) {
  return `Student: ${name}`;
}

export function calculateGrade(marks) {
  if (marks >= 80) return "A";
  else if (marks >= 60) return "B";
  else if (marks >= 0) return "C";
  return "Invalid marks";
}

// app.mjs
import studentInfo, { calculateGrade } from "./result.mjs";
console.log(studentInfo("Junaid"));
console.log(`Grade: ${calculateGrade(80)}`); // A
console.log(`Grade: ${calculateGrade(70)}`); // B
console.log(`Grade: ${calculateGrade(45)}`); // C
```

> **Return vs console.log**
>
> return function se value wapas bhejta hai. console.log() sirf console par print karta hai aur khud undefined return karta hai. Reusable functions mein result return karo; app.mjs mein print karo.

Boundary test

Boundary check: 80 - A; 79 - B; 60 - B; 45 - C. `>=` mein equality include hoti hai.

03 | HTTP Module - Own Server

Node.js ka built-in http module use kar ke server ban sakta hai. req request ko represent karta hai, aur res response ko.



Working server: routing, HTML aur real 404

```
import http from "node:http"; // ES Module version

const server = http.createServer((req, res) => {
  res.setHeader("Content-Type", "text/html");

  if (req.url === "/") {
    res.statusCode = 200;
    res.end("<h1>Home Page</h1><p>Welcome!</p>");
  } else if (req.url === "/courses") {
    res.statusCode = 200;
    res.end("<h1>Courses</h1><p>Learning Node.js</p>");
  } else {
    res.statusCode = 404;
    res.end("<h1>404</h1><p>Page Not Found</p>");
  }
});

server.listen(8000, () => {
  console.log("Server running at http://localhost:8000");
});
```

Agar tum CommonJS use kar rahe ho to first line ki jagah const http = require("http"); bhi sahi hai (CommonJS file mein).

HTTP mein yaad rakhne wale rules

Method / property	Purpose
res.setHeader("Content-Type", "text/html")	Browser ko HTML content type batata hai.
res.statusCode = 404	Unknown route par real Not Found status.
res.end("...")	Response send / complete karta hai.
if / else if / else	Exactly one matching route or fallback.

04 | EventEmitter - Event Handling

Node.js ka events module actions (events) aur unke listeners se kaam karta hai. .on() ek listener register karta hai, .emit() event fire karta hai.



Important methods

Method	Kya karta hai?
.on(event, fn)	Har emission par callback run hota hai.
.once(event, fn)	Sirf pehli emission par run hota hai.
.emit(event, data)	Event trigger karta hai aur arguments pass karta hai.
.off(event, fn)	Specific listener remove karta hai.

Practical example: 2 listeners, phir 1 remove

```
import { EventEmitter } from "node:events";
const emitter = new EventEmitter();

function sendSMS(amount) {
  console.log(`SMS: ${amount} PKR`);
}
function saveLog(amount) {
  console.log(`Log: ${amount} PKR`);
}

emitter.once("transaction", () => console.log("Started"));
emitter.on("transaction", sendSMS);
emitter.on("transaction", saveLog);

emitter.emit("transaction", 5000); // Started, SMS, Log
emitter.off("transaction", sendSMS);
emitter.emit("transaction", 10000); // Only Log
```

Common errors

on("x", sendSMS) sahi function reference hai; on("x", sendSMS(5000)) ghalat hai agar listener register karna ho. Event names case-sensitive hain: "welcome" aur "Welcome" alag events hain.

05 | All Important Tips + Final Revision

Common mistake	Correct rule
"Context-Type" header	Hamesha "Content-Type" likho.
HTTP 400 for page not found	Unknown URL ke liye 404 set karo.
Status code after res.write	Headers/status hamesha body se pehle.
req.end() on server	Server response complete karne ke liye res.end().
Independent if + else	Mutually exclusive routes: if / else if / else.
import "./math.js" but file math.mjs	Exact file name / extension match.
return console.log(...)	return value; calling file mein console.log.
marks => 60	Correct comparison marks >= 60.
marks >= 60 || marks <= 79	Range mein && use karo, ya ordered conditions.
.off without callback	Same named function reference pass karo.

Quick revision questions

Q1: Named import mein braces kab use hoti hain? Q2: Default export import ke waqt braces lagti hain? Q3: Marks 80 par A ke liye kaunsa operator? Q4: Unknown route ke liye status 400 ya 404? Q5: .off() ko remove karne ke liye kya dena hota hai?

> **Short answers**
>
> Q1: named import mein { } | Q2: nahi | Q3: >= | Q4: 404 | Q5: same listener function reference.

Next learning step

Ab tum ES Modules basics ke baad practice mein multiple-file utility modules, ESM se built-in Node modules import, aur HTTP mein JSON response / GET-POST methods seekh sakte ho. Express.js tab start karo jab basic HTTP concepts comfortable hon.

Completed practice record: EventEmitter 56/90 (9 tasks); HTTP 50/70 (7 tasks); ES Modules 30/40 (4 tasks), final logic correction successful.

Ye notes tumhari recent chat practice par based hain. Examples dobara run kar ke output verify karna achhi coding habit hai.